



Evaluation of pathological alpha-synuclein in skin biopsies as potential biomarker for Parkinson's disease

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Summary

Biomarkers to detect and quantify pathological alpha-synuclein (aSyn) in accessible samples from Parkinson's disease (PD) patients are urgently needed. We have developed two different assays to measure pathological aSyn in the skin of PD cases: aSyn seed amplification assay (SAA) and a ligand-binding assay (LBA) for aSyn aggregates phosphorylated at serine 129 (pS129). Here, we measured 120 skin biopsies from prodromal, sporadic PD, and control individuals from the PPMI cohort. The skin biopsies were first homogenized using a bead ruptor and then measured by SAA and LBA. Our primary aim of this study is to determine the diagnostic accuracy of both methods.

Method

Homogenization of skin biopsies

The weight of each frozen skin biopsy was determined. For homogenization, 500 μ L tubes were used and each tube was filled with ten yttria-stabilized zirconium oxide beads (MP Biomedicals, California, United States). Ten-fold volume (v/w) of lysis buffer (1x TBS supplemented with protease and phosphatase inhibitors (Roche) and 0.5 % Triton X100) was added to the tubes, e.g. 100 μ L lysis buffer for processing of a 10 mg biopsy. The tubes were then placed on wet ice. Biopsies were processed in batches of ten. Each biopsy was first smashed with a hammer while it was still frozen and then transferred to a 500 μ L tube containing beads and lysis buffer. The biopsies were homogenized with a Bead Ruptor (Omni International, Georgia, United States) in six 10 s cycles at 5.5 m/s. Samples were removed from the Bead Ruptor after each 10 s cycle and placed for 1 min on wet ice before the next cycle was started. After the last cycle, the samples were placed back on wet ice. The samples were centrifuged for 1 min at 2000 xg (4°C) and as much of the liquid as possible was collected into a new tube. To remove debris and collagen fibers the samples were centrifuged for 20 min at 21,000 xg (4 °C). The supernatant was transferred into a new tube, aliquoted, and frozen on dry ice. The pellet was discarded. Samples were stored at -80 °C until further use.

pS129-aSyn aggregate LBA

The assay was run on the Singulex Erenna platform (Merck, Darmstadt, Germany) using aSyn aggregate selective capture and pS129 aSyn selective detection antibodies. Capture and detection





antibodies were labelled using Capture/Detection Labeling Kits according to manufacturer's instructions (Merck, Darmstadt, Germany). Magnetic beads coupled with capture antibodies were prepared in Coated Bead Buffer at 10 mg/mL. Detection antibody was adjusted to 1 mg/mL during labeling with fluorescent dye. Skin lysate samples (1:5 dilution) and calibrators were diluted in Standard Diluent (Merck, Darmstadt, Germany) and run in two technical replicates. One hundred μ L of diluted sample or calibrator was added per well of a 96-well plate (Corning, Reynosa, Mexico). Capture beads were diluted 1:200 in Assay/Discovery buffer (Merck, Darmstadt, Germany) and 100 μ L/well was added (5 μ g beads/well). The plate was incubated for 2 h at 25 °C in a shaker. Beads were then washed with Erenna System buffer (Merck, Darmstadt, Germany) and resuspended with 20 μ L/well of 550 ng/mL detection antibody in Assay/Discovery buffer. The plate was incubated for 1 h at 25 °C in a shaker. After washing, the beads were resuspended and transferred into a new 96-well plate. The wash buffer was then removed from beads in a final aspiration step. A total of 9.5 μ L/well of Elution buffer B (Merck, Darmstadt, Germany) was added to the beads and incubated for 20 min at 25 °C in a shaker. The eluate was magnetically separated from the beads, transferred into a 384-well plate (Corning, Reynosa, Mexico) and neutralized immediately with 9.5 μ L/well of Buffer D (Merck, Darmstadt, Germany). After heat-sealing, the plate was analyzed using the Singulex Erenna instrument. The dilution corrected limit of detection (LOD) for skin lysates was estimated as 2 pg/mL. If no value was calculated by the analysis software (signal at blank level) the value was set to 0.67 pg/mL (1/3 of LOD). Values that were calculated by the software and that were between 0.67 and 2 pg/mL were reported in the results file. Samples with one replicate below LOD and the other replicate at least 2-fold above LOD were retested in a second assay run. For such samples the results of the second assay run were reported in the results file.

aSyn SAA

Human recombinant full-length (1-140 aa) alpha-synuclein without tag was expressed in *E. coli* and purified using standard purification methods. Alpha-synuclein from one single purification batch was used throughout this study. Directly before use in SAA, alpha-synuclein was centrifuged for 30 min at 21,000 xg (4°C). A mastermix containing 44.4 mM sodium phosphate (pH 8.2), 0.0167 % Triton X-100, 11.1 μ M Thioflavin T (ThT), and 0.11 mg/ml alpha-synuclein was prepared. To each well of a black 384-well plate (Greiner bio-one, Germany), 45 μ L of mastermix and 6.5 ± 1 mg of 0.1 mm zirconia/silica beads (Roth, Karlsruhe, Germany) were added. Skin samples were tested in three different dilutions with 4 technical replicates each. Five μ L of pre-diluted skin lysate (pre-dilution in 100 mM sodium phosphate (pH 8.2)) were added per well yielding a final reaction volume of 50 μ L/well and final skin lysate dilutions of 1:400, 1:800, and 1:1600. The final reaction mixture (50 μ L/well) contained 50 mM sodium phosphate (pH 8.2), 0.015 % Triton X-100, 10 μ M ThT, and 0.1 mg/ml alpha-synuclein. The plates were sealed with a plate sealer film (Fisher Scientific Ltd, UK) and incubated in a BMG FluoSTAR Omega plate reader at 32 °C for 70 h with shake/rest cycles: 1 min double orbital shaking at 200 rpm, 14 min rest. ThT fluorescence measurements (450 nm excitation and 480 nm emission; top read) were taken every 15 min.





Before any further analysis, each aggregation curve was baseline corrected by subtracting the mean of the relative fluorescence units (rfu) measured between 2 and 5 h from all data points of the curve. A threshold of 3500 rfu was used to determine the time to threshold (TTT), i.e. the time when the curve reaches this threshold. If the threshold was not reached during the 70 h runtime of the assay, the TTT was set to 70 h. Curves that reached the threshold were defined as positive replicates. A sample was defined as SAA positive if 3 or 4 replicates reached the threshold within 70 h. If a sample reached the threshold in 2 out of 4 replicates, then the run was repeated for this sample. For such samples, the results of the second assay run were reported in the results file. If the second assay run showed again 2 out of 4 positive replicates, then the sample was rated as positive. In addition, the maximum fluorescence (Fmax) was determined as the highest rfu value recorded during the 70 h runtime of the assay. The area under curve (AUC) was calculated between 0 and 70 h. Negative AUC values that can occur due to minimal baseline drift were set to 0.

Table 1: Overview of the terms used in the results file.

TESTNAME	Description
SAA_1:400_status SAA_1:800_status SAA_1:1600_status	Final result indicating if a skin sample was SAA positive (1) or negative (0) based on 4 technical replicates.
SAA_1:400_TTT (rep 1-4) SAA_1:800_TTT (rep 1-4) SAA_1:1600_TTT (rep 1-4)	Time to threshold (TTT) in h for every sample and for each of the 4 technical replicates.
SAA_1:400_Fmax (rep 1-4) SAA_1:800_Fmax (rep 1-4) SAA_1:1600_Fmax (rep 1-4)	Maximum fluorescence (Fmax) in rfu for every sample and for each of the 4 technical replicates.
SAA_1:400_AUC (rep 1-4) SAA_1:800_AUC (rep 1-4) SAA_1:1600_AUC (rep 1-4)	Area under the curve (AUC) for every sample and for each of the 4 technical replicates.
LBA (rep 1-2)	Ligand-binding assay (LBA) quantification of pS129 aSyn aggregates in pg/mL for 2 technical replicates.

About the Authors

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